

|                                                    |                                                 |
|----------------------------------------------------|-------------------------------------------------|
| Program : <b>Diploma in Biomedical Engineering</b> |                                                 |
| Course Code : <b>6242C</b>                         | Course Title: <b>Biomedical Instrumentation</b> |
| Semester : <b>6</b>                                | Credits: <b>4</b>                               |
| Course Category : <b>Open Elective</b>             |                                                 |
| Periods per week: <b>4 (L:4 T:0 P:0)</b>           | Periods per semester : <b>60</b>                |

Course Objectives:

- To impart knowledge about the principle and working of different types of biomedical equipment.
- To introduce the signal characteristics of various body signals.

Course Prerequisites: Nil

Course Outcomes:

On completion of the course, the student will be able to:

| CO <sub>n</sub> | Description                                                                                                                                      | Duration (Hours) | Cognitive Level |
|-----------------|--------------------------------------------------------------------------------------------------------------------------------------------------|------------------|-----------------|
| CO1             | Illustrate various body signals and describe the signal acquisition systems viz ECG, EEG and EMG                                                 | 15               | Understanding   |
| CO2             | Describe the functioning of various therapeutic equipments viz. defibrillator, pacemaker, surgical diathermy, ventilator and heart lung machine. | 14               | Understanding   |
| CO3             | Explain the working principle and block diagram of medical imaging machines.                                                                     | 15               | Understanding   |
| CO4             | Identify an analytical instrument for a specified application by knowing its working and application.                                            | 14               | Understanding   |
|                 | Series Test                                                                                                                                      | 2                |                 |

CO-PO Mapping:

| Course Outcomes | PO1 | PO2 | PO3 | PO4 | PO5 | PO6 | PO7 |
|-----------------|-----|-----|-----|-----|-----|-----|-----|
| CO1             | 2   |     |     |     |     |     |     |
| CO2             | 2   |     |     |     |     |     |     |
| CO3             | 2   |     |     |     |     |     |     |
| CO4             | 2   |     |     |     |     |     |     |

3-Strongly mapped, 2-Moderately mapped, 1-Weakly mapped

#### Course Outline:

| Module Outcomes                                                                                                                                                                                                                                                                                                                                                                                                     | Description                                                                                                                                             | Duration (Hours) | Cognitive Level |
|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------|------------------|-----------------|
| CO1                                                                                                                                                                                                                                                                                                                                                                                                                 | <b>Illustrate various body signals and describe the signal acquisition systems viz ECG, EEG and EMG</b>                                                 |                  |                 |
| M1.01                                                                                                                                                                                                                                                                                                                                                                                                               | Describe the basics of resting membrane potential and action potential                                                                                  | 4                | Understanding   |
| M1.02                                                                                                                                                                                                                                                                                                                                                                                                               | Illustrate the ECG waveform and its acquisition                                                                                                         | 5                | Understanding   |
| M1.03                                                                                                                                                                                                                                                                                                                                                                                                               | List types of brain waves and its significance and explain the working of EEG machine                                                                   | 3                | Understanding   |
| M1.04                                                                                                                                                                                                                                                                                                                                                                                                               | Define body signals- EMG, ERG , EOG                                                                                                                     | 3                | Understanding   |
| <b>Contents:</b><br>Resting membrane potential in a nerve cell, action potential, propagation of action potential, recording of action potential.<br>ECG waveform, Bipolar lead system in ECG, Block diagram of a single channel ECG machine, Holter monitor<br>EEG waveforms and its characteristics, 10-20 electrode system, Block diagram of EEG machine<br>Signal characteristics and concepts of EMG, ERG, EOG |                                                                                                                                                         |                  |                 |
| CO2                                                                                                                                                                                                                                                                                                                                                                                                                 | <b>Describe the functioning of various therapeutic equipments viz. defibrillator, pacemaker, surgical diathermy, ventilator and heart lung machine.</b> |                  |                 |
| M2.01                                                                                                                                                                                                                                                                                                                                                                                                               | Explain the working of defibrillator and pacemaker                                                                                                      | 6                | Understanding   |
| M2.02                                                                                                                                                                                                                                                                                                                                                                                                               | Describe the functioning of a surgical diathermy                                                                                                        | 2                | Understanding   |
| M2.03                                                                                                                                                                                                                                                                                                                                                                                                               | Explain the need and working of basic ventilator                                                                                                        | 3                | Understanding   |
| M2.04                                                                                                                                                                                                                                                                                                                                                                                                               | Explain the working of lithotripsy and heart lung machine                                                                                               | 3                | Understanding   |
|                                                                                                                                                                                                                                                                                                                                                                                                                     | Series Test-I                                                                                                                                           | 1                |                 |

**Contents:**

Fibrillation, Need for defibrillator, types of defibrillators, Block diagram of a external defibrillator, Operation of AED

Natural pacemaker of body, need for pacemaker, types of pacemakers - external, internal, programmable. Block diagram of a programmable pacemaker

Surgical diathermy - purpose, modes; basic block diagram of an ESU

Ventilator - purpose, types of ventilators, modes of operation, Block diagram of a microprocessor based ventilator.

Lithotripsy - principles, types & applications. Heart lung machine - need and block diagram

|            |                                                                                     |   |               |
|------------|-------------------------------------------------------------------------------------|---|---------------|
| <b>CO3</b> | <b>Explain the working principle and block diagram of medical imaging machines.</b> |   |               |
| M3.01      | Explain the working of X-ray machines                                               | 4 | Understanding |
| M3.02      | Describe the working principle and block diagram of CT machine                      | 4 | Understanding |
| M3.03      | Illustrate the principle of operation and block diagram of MRI machine              | 4 | Understanding |
| M3.04      | Explain the ultrasound imaging modality                                             | 3 | Understanding |

**Contents:**

X-rays - properties and generation, block diagram of X-ray machine

CT machine - working principle- linear attenuation coefficient, CT number, Block diagram of CT machine

MRI machine - principle - FID signal, T1 and T2 relaxation time. Block diagram

Ultrasound machine - properties of ultrasound, acoustic impedance, modes of operation, Basic pulse echo machine block diagram

|            |                                                                                                              |   |               |
|------------|--------------------------------------------------------------------------------------------------------------|---|---------------|
| <b>CO4</b> | <b>Identify an analytical instrument for a specified application by knowing its working and application.</b> |   |               |
| M4.01      | Describe the working of spectrophotometers                                                                   | 4 | Understanding |
| M4.02      | Illustrate the working and block diagram of colorimeter, flame photometer and pH meter                       | 6 | Understanding |
| M4.03      | Explain the working of blood gas analyzers and coulter counters                                              | 4 | Understanding |
|            | Series Test - II                                                                                             | 1 |               |

**Contents:**

Define Beer Lambert's law, list types of spectrophotometers , working of UV spectrometer

Colorimeter - principle of operation and block diagram; flame photometer - principle of operation, block diagram and application; pH meter - principle and block diagram

Blood gas analyzers - need, principle and principle of operation

Blood cell counters - coulter counter - principle and block diagram

**Text / Reference:**

| <b>T/R</b> | <b>Book<br/>Title/Author</b>                                                                                               |
|------------|----------------------------------------------------------------------------------------------------------------------------|
| T1         | Carr & Brown, Introduction to Biomedical Equipment Technology, 4 <sup>th</sup> edition, Pearson Education, Singapore, 2001 |
| T2         | R S Kandpur, Hand book of Biomedical instrumentation, Tata McGraw Hill 3 <sup>rd</sup> edition, 2014                       |
| R1         | Leslie Cromwell, Fred J. Weibell, Erich A. Pfeiffer, Biomedical Instrumentation and Measurements, PHI, 2nd Edition, 2004   |
| R2         | R S Kandpur, Hand book of Analytical Instruments, McGraw Hill Education Pvt Limited, 2006                                  |

**Online resources:**

| <b>Sl.No</b> | <b>Website Link</b>                                                                                                                                                                                                     |
|--------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1            | <a href="https://en.ecgpedia.org/wiki/Basics">https://en.ecgpedia.org/wiki/Basics</a>                                                                                                                                   |
| 2            | <a href="https://en.wikipedia.org/wiki/Defibrillation">https://en.wikipedia.org/wiki/Defibrillation</a>                                                                                                                 |
| 3            | <a href="https://www.allaboutcircuits.com/technical-articles/engineer-introduction-to-mechanical-ventilation/">https://www.allaboutcircuits.com/technical-articles/engineer-introduction-to-mechanical-ventilation/</a> |
| 4            | <a href="https://radiologykey.com/imaging-equipment/">https://radiologykey.com/imaging-equipment/</a>                                                                                                                   |
| 5            | <a href="https://microbenotes.com/spectrophotometer-principle-instrumentation-applications/">https://microbenotes.com/spectrophotometer-principle-instrumentation-applications/</a>                                     |